

Data Pipelines For Pretraining

AI Engineering From Scratch #10

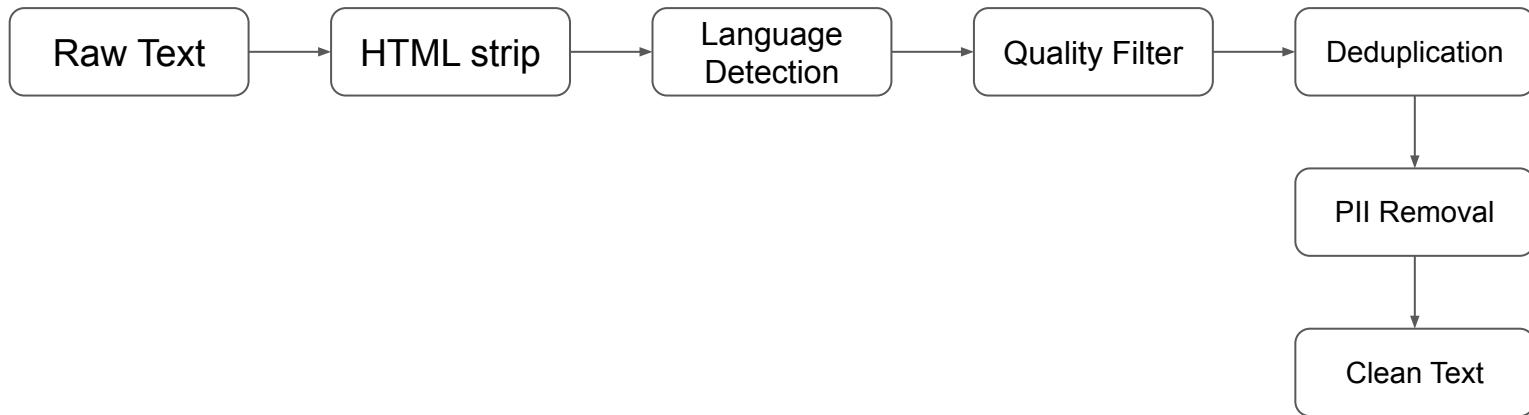
Where the data comes from ?

Every large language model is trained on a mix of sources:

- Common crawl
- Wikipedia
- Github code
- Books (BookCorpus, Pile)
- Academic Paper (arxiv, S2ORC)
- StackOverflow, Reddit
- Curated web (C4, RefinedWeb)

Data Cleaning

A typical pipeline for Common Crawl data cleaning:



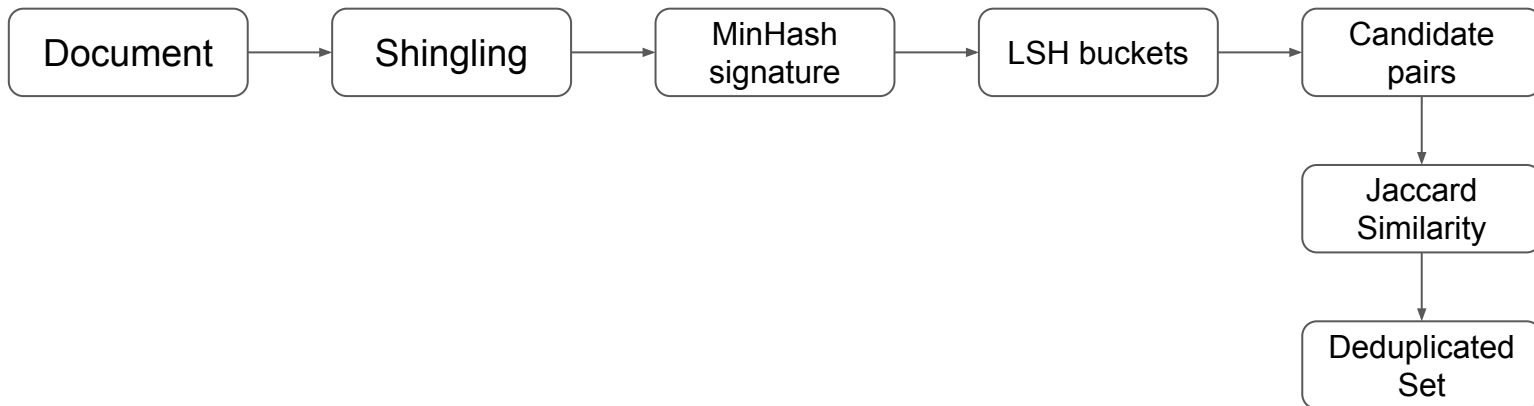
Data Cleaning

*NOTE:

- PII removal: Names, email addresses, phone numbers, social security number. Regex-based detection for structured PII, NER models for names in context.

Deduplication with MinHash

- Exact deduplication is easy: hash each document, remove duplicates.
- But near-duplicates are the real problem:
 - Two copies of the same new article with slightly different ads around it are near duplicates.
 - The content is 95% identical, but byte-for-byte they differ.
- MinHash and Locality-Sensitive Hashing (LSH) solves this efficiently:



Deduplication with MinHash

- **Shingling:** Convert document into set of n-grams. Example: “the quick brown fox” with 3-word shingles becomes {“the quick brown”, “quick brown fox”}.
- **MinHash:** For each document’s shingle set, compute k hash values. Each hash value is the minimum hash across all shingles under a different hash function. This creates a fixed-size “signature” that approximates the Jaccard similarity between any two documents.
- **LSH:** Group document into buckets based on bands of their MinHash signature. Documents on the same bucket are candidate near-duplicates. This avoid comparing every pair - you only compare candidates.

Deduplication with MinHash

- **Verify:** For each candidate pair, compute exact Jaccard similarity. Remove a copy if similarity exceed a threshold (typically 0.8).

Sequence Packing

- Pack multiple documents into a single sequence, separated by EOS tokens.
- The attention mask must be set correctly. Tokens from Document A should not attend to tokens from Document B within the same packed sequence. This requires block diagonal attention mask.